# CCI Momentum Strategy

## Backtest, Analysis & Notebook

*S&P 500 | 2000–2025 | 25-Year Analysis*

A comprehensive guide to the 9-period Commodity Channel Index momentum strategy — covering entry logic, risk metrics, Monte Carlo simulation, and professional execution insights.

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# Introduction to the CCI Momentum Strategy

## What is the CCI Momentum Strategy?

The CCI Momentum Strategy is built on the Commodity Channel Index (CCI), a momentum-based oscillator developed by Donald Lambert that measures the relationship between an asset's current price and its historical average price. Unlike many bounded oscillators, the CCI is unbounded — meaning it can theoretically rise or fall without limit, making it particularly suited to detecting short-term price extremes.

When the CCI is above zero, the price is trading above its historical average, signalling relative strength. When below zero, the price is below its historical average, signalling potential weakness or oversold conditions. This strategy exploits the latter — entering long positions when the CCI signals an oversold extreme, then exiting as the price recovers past the prior day's high.

## The Core Concept

| Condition | Action |
|---|---|
| CCI(9) drops below −30 | BUY — Market is in an oversold condition |
| Close crosses above yesterday's high | SELL — Momentum has reversed; take profit |

> *Think of CCI below −30 as a "rubber band stretched too far" below the mean. The strategy bets on the snap-back — the tendency for price to revert toward its average after an extreme.*

## Why CCI(9)?

The 9-period lookback strikes a balance between sensitivity and noise. A shorter period would generate signals on every minor intraday fluctuation; a longer period would be too slow to capture the short mean-reversion windows that define this strategy's edge. At -30, the threshold is deliberately shallow — capturing frequent oversold events while managing against chronic trending weakness.

> *Note: Because the CCI threshold (-30) and period (9) are both relatively short-term parameters, the strategy is inherently susceptible to overfitting. Results should be interpreted with this risk in mind.*

## Entry Logic

| Step | Condition |
|---|---|
| 1 | Monitor the S&P 500 (or chosen asset) daily |
| 2 | Calculate the 9-period CCI on the daily close |
| 3 | If CCI(9) < −30 → Enter a LONG position at next day's open |
| 4 | If CCI(9) ≥ −30 → No action; remain in cash |

## Exit Logic

| Step | Condition |
|---|---|
| 1 | Continue monitoring the position daily |
| 2 | Check if today's close exceeds yesterday's high |
| 3 | If YES → Exit the position at the close |
| 4 | If NO → Hold position; await the recovery signal |

## Key Definitions

| Term | Definition |
|---|---|
| CCI (Commodity Channel Index) | An oscillator measuring the deviation of price from its statistical mean |
| 9-period CCI | CCI calculated over the last 9 trading days |
| Oversold Level (−30) | The entry threshold where price is deemed abnormally cheap vs. its mean |
| Yesterday's High | The prior session's intraday high; breaching it on the close confirms recovery |
| Mean Reversion | The tendency for a price that has moved far from its average to return toward it |
| Drawdown | Peak-to-trough decline in account equity |
| Whipsaw | Entry followed immediately by continued decline before recovery |

## How This Strategy Works

### 1. Captures Short-Term Oversold Bounces

The S&P 500, despite its long-term upward bias, experiences frequent short-term pullbacks. The CCI(9) below -30 identifies these windows efficiently, allowing the strategy to enter at statistically depressed prices with a high probability of near-term recovery.

### 2. Objective and Rules-Based

There is no discretion involved. The CCI threshold is fixed, the exit rule is fixed, and the instrument is fixed. This eliminates emotional bias — the algorithm executes mechanically on every qualifying signal.

### 3. Frequent Trading, Short Duration

With 583 trades over 25 years (approximately 22.5 per year), the strategy is far more active than trend-following alternatives. The average holding period is just 5.3 days — capital is typically deployed briefly and returned to cash quickly.

## Pros and Cons

| Benefit | Cost |
|---|---|
| High win rate (~72%) provides psychological comfort | Small average wins vs. larger losses (inverted payoff ratio) |
| Frequent signals keep capital actively working | Susceptibility to overfitting on short-term parameters |
| Mean reversion logic is well-supported in literature | Performance can degrade during strong trending markets |
| No stop-loss / no profit target — simplicity | Requires full-size exposure with no downside protection |

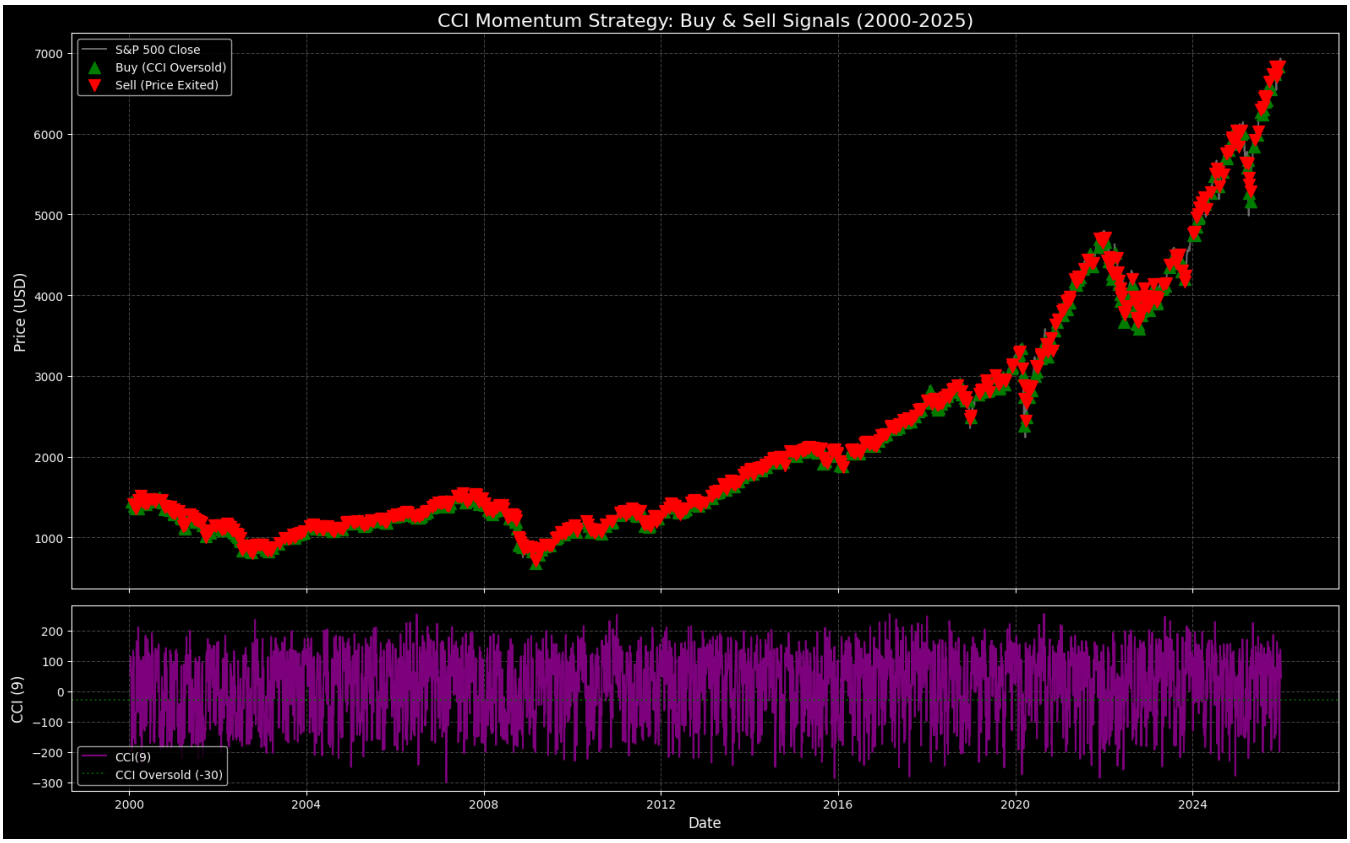
## Important Warnings

### ⚠ This is a Mean-Reversion Strategy — NOT Trend-Following

| Expectation | Reality |
|---|---|
| Win 50–60% of trades | You win ~72% — but average winners are smaller than average losers |
| Large, compounding gains | Returns are incremental; profit accumulates through frequency, not magnitude |
| Crash protection built-in | No stop-loss means full exposure during rare but severe adverse moves |

> *Warning: The strategy's payoff ratio (avg win / avg loss) is below 1. It profits through win rate advantage, not through size of winners. One cluster of consecutive losses can significantly damage the equity curve.*

**SECTION SUMMARY** *The CCI Momentum Strategy enters long when the 9-period CCI drops below -30 (oversold extreme) and exits when the close recovers above yesterday's high. It is a high-frequency, short-duration mean-reversion system that profits through win-rate advantage rather than large payoffs. Its key risks are overfitting susceptibility, inverted payoff ratio, and the absence of a hard stop-loss.*


CCI Momentum Strategy: Buy & Sell Signals (2000-2025)
S&P 500 Close
Buy (CCI Oversold)
Sell (Price Exited)
Price (USD)
7000
6000
5000
4000
3000
2000
1000
CCI (9)
200
100
0
−100
−200
−300
CCI(9)
CCI Oversold (-30)
2000
2004
2008
2012
2016
2020
2024
Date

# Executive Summary: The High-Frequency Mean Reversion Edge

## Performance Summary: CCI Momentum Strategy (2000–2025)

The CCI Momentum Strategy presents a contrasting profile to conventional trend-following systems. It is high-frequency, short-duration, and wins most of the time — but the average win is smaller than the average loss. Its value proposition rests entirely on frequency: consistently being right more than two-thirds of the time, and keeping losers sufficiently rare to preserve cumulative returns.

| Category / Metric | Value | Professional Interpretation |
|---|---|---|
| CAGR | 8.80% | Strong absolute return; competitive with passive S&P 500 over the period. |
| Profit Factor | 1.71 | Solid. For every $1.00 lost, the system returned $1.71 in gains. |
| Max Drawdown | -16.49% | Excellent result. The worst peak-to-trough decline stayed under 17%. |
| Annualized Vol | 8.74% | Low volatility profile; significantly calmer than buy-and-hold S&P 500. |
| Sharpe Ratio | 0.49 | Decent risk-adjusted return; respectable for a long-only equity system. |
| Win Rate | 71.87% | High win rate provides operational consistency; ~3 winners for every loser. |
| Avg Trade PnL | 0.40% | Small per-trade profits; system relies on compounding frequency. |
| Num Trades | 583 | High sample size over 25 years (∮22.5 per year); statistically robust. |

**SECTION SUMMARY** *The CCI Momentum Strategy delivers 8.80% CAGR with a controlled -16.49% max drawdown and a 71.87% win rate over 583 trades. The trade-off: average wins (1.34%) are smaller than average losses (-1.99%), requiring the high win rate to maintain positive expectancy. The system's edge is frequency and discipline, not magnitude.*

## Comprehensive Performance Metrics

| Metric | Value | USD$ |
|---|---|---|
| Total Net Profit | 788.44% | $788,438.23 |
| Gross Profit | 560.15% | $560,151.32 |
| Gross Loss | -327.03% | -$327,030.19 |
| Profit Factor | 1.71 | N/A |
| Max. Drawdown (Equity) | -16.49% | N/A |
| Sharpe Ratio | 0.49 | N/A |
| Sortino Ratio | 0.49 | N/A |
| Ulcer Index | 0.0392 | N/A |
| Kelly Criterion | 29.91% | N/A |
| Total # of Trades | 583 | N/A |
| Percent Profitable | 71.87% | N/A |
| # of Winning Trades | 419 | N/A |
| # of Losing Trades | 164 | N/A |
| # of Even Trades | 0 | N/A |
| Avg. Trade PnL | 0.40% | $1,352.41 |
| Avg. Winning Trade PnL | 1.34% | $1,337.83 |
| Avg. Losing Trade PnL | -1.99% | -$1,994.08 |
| Ratio Avg. Win / Avg. Loss | 0.67 | N/A |
| Max. Consec. Winners | 14 | N/A |
| Max. Consec. Losers | 4 | N/A |
| Largest Winning Trade PnL | 8.54% | $8,543.40 |
| Largest Losing Trade PnL | -14.50% | -$14,497.30 |
| Avg. # of Trades per Day | 0.0893 | N/A |
| Avg. Time in Market | 5.32 days | N/A |
| Profit per Month | 2.54% | $2,535.72 |
| Avg. MAE | -2.08% | N/A |
| Avg. MFE | 1.64% | N/A |
| Avg. ETD | 1.24% | N/A |

## The Mathematical Engine: Win-Rate Driven Expectancy

Unlike trend-following strategies that win rarely but win big, the CCI Momentum Strategy wins frequently but with modest individual gains. This is a fundamentally different statistical engine.

### The Formula

> *Expectancy = (Win % × Average Win) − (Loss % × Average Loss)*

| Component | Value |
|---|---|
| Win Rate | 71.87% |
| Loss Rate | 28.13% |
| Average Winning Trade | +1.34% |
| Average Losing Trade | -1.99% |
| Net Expectancy per Trade | +0.40% |

Even though the average loss is larger than the average win, the strategy maintains positive expectancy because winners outnumber losers by more than 2.5 to 1. This is the mathematical heart of high-frequency mean-reversion: volume and consistency compensate for inverted payoff ratios.

## The Systematic Advantage: Patience Over Prediction

The CCI Momentum Strategy does not predict where the market is going. It identifies when the market has temporarily gone too far in one direction and bets on a short-term reversion. This makes it emotionally challenging for a human discretionary trader, because:

- Trades are entered during drawdowns, when fear is highest.
- Exits are mechanically triggered, removing the temptation to hold for more.
- There is no stop-loss — the trader must tolerate open adverse movement.

| Advantage | Description |
|---|---|
| Consistency | The algorithm enters every qualifying CCI signal without hesitation, fatigue, or fear. |
| Discipline | It exits when the rule says exit — not when the trader “feels” the move is over. |
| Speed | Positions are held for an average of 5.3 days. Capital recycling is rapid. |

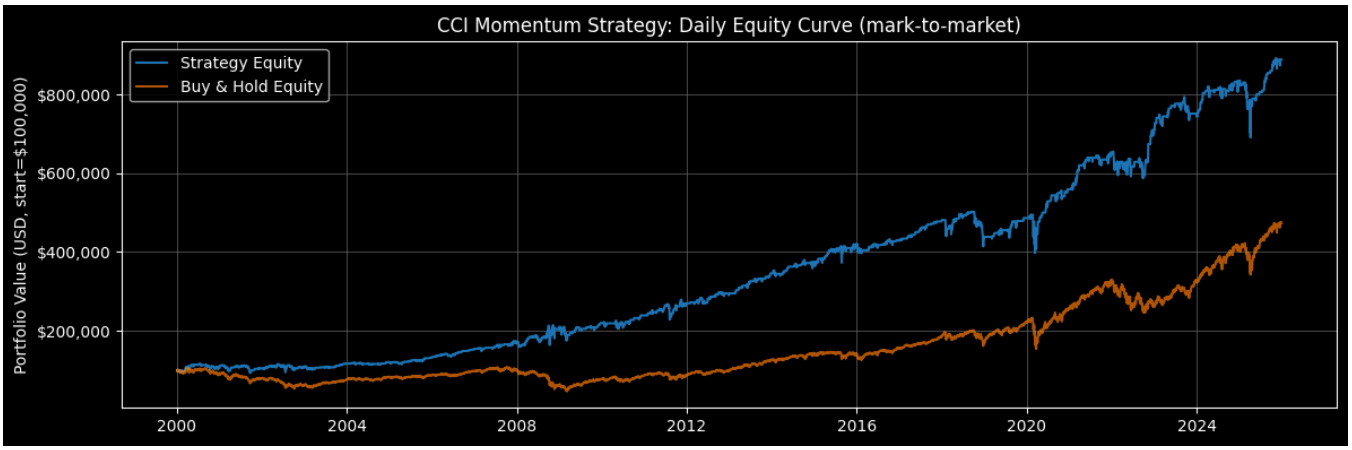
## Strategy Identity: The “High-Frequency Reversion Engine”

| Metric | Value |
|---|---|
| CAGR | 8.80% |
| Max Drawdown | -16.49% |
| Win Rate | 71.87% |

**SECTION SUMMARY** *The strategy's positive expectancy is driven by a win-rate of 71.87% and a per-trade expectancy of +0.40%, despite an inverted win/loss ratio of 0.67. It profits through consistent frequency and disciplined mechanical execution — not through large individual winners.*

# The Daily Equity Curve

The Daily Equity Curve provides a direct comparison between the CCI Momentum Strategy and a simple Buy & Hold approach for the S&P 500. This is the most intuitive view of strategy performance — it shows the dollar value of the account on every trading day, exactly as it would appear in a brokerage statement.




## The "Smooth Compounder" Pattern

The most immediate observation is that the strategy's equity curve grows more steadily than the raw S&P 500 Buy & Hold curve. Because the strategy holds positions for only 5.3 days on average, the majority of the time is spent in cash — insulating the portfolio from sustained trending drawdowns.

## Crisis Period Behaviour

| Crisis Period | Buy & Hold (B&H) |
|---|---|
| 2000–2002 Dot-com Crash | >-50% collapse |
| 2008 Financial Crisis | >-50% collapse |
| 2020 COVID Crash | >-34% in weeks |
| 2022 Bear Market | ~-19% |

The strategy is not a perfect hedge — it takes long positions, so sustained downtrends can still cause losses. However, because positions are short-duration, losses during crises tend to be shallower than in Buy & Hold.

## The Bull Market Relationship

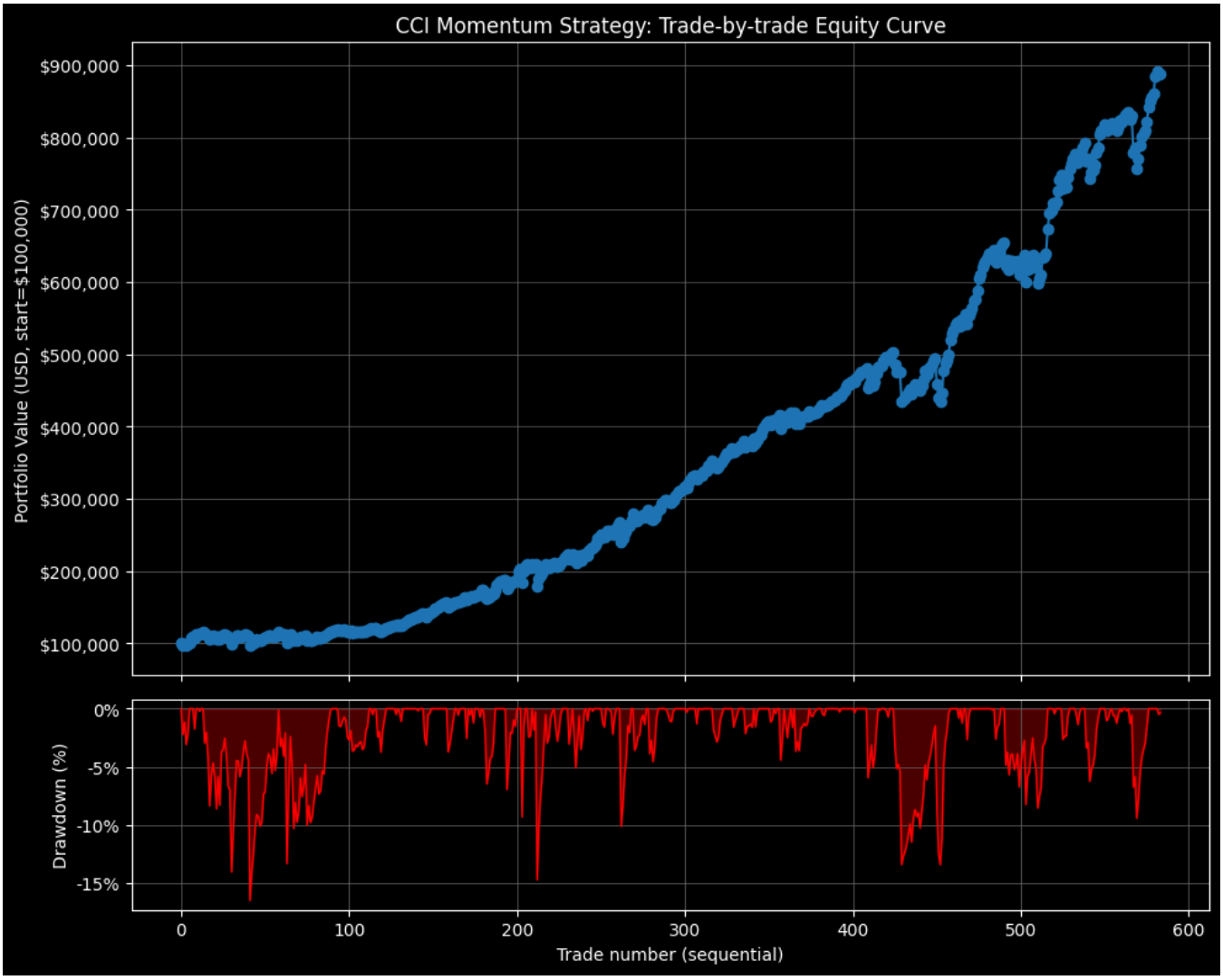
During strong uninterrupted bull markets, the strategy can underperform Buy & Hold. Since it only enters after a CCI oversold reading, it misses the early momentum of rallies. The trade-off: in exchange for missing some upside, it avoids the catastrophic 40–60% drawdowns that define major market corrections.

| Metric | Value |
|---|---|
| Sharpe Ratio | 0.49 |
| Annualized Volatility | 8.74% |

**SECTION SUMMARY** *The Daily Equity Curve shows a steady, compounding uptrend with shallow drawdowns. The strategy avoids the catastrophic crashes that define Buy & Hold risk, but it does not perfectly hedge — it is still a long-only system. Its advantage is the short holding period, which naturally limits exposure during broad market downtrends.*

# Trade-by-Trade Equity Curve & Drawdown

While the Daily Equity Curve shows account value over calendar time, the Trade-by-Trade Equity Curve reveals the strategy's "career" as a series of discrete decision events. Each data point represents the completion of one trade — making it the purest view of the strategy's statistical engine.




## Trade-by-Trade Analysis: The “Career” of a Mean Reversion System

| Metric | Value |
|---|---|
| Total Trades | 583 |
| Max Consecutive Losers | 4 |
| Max Consecutive Winners | 14 |
| Profit Contribution | ∮72% of trades (winners) |

| | |
|---|---|
| Starting Capital | $100,000 |
| Final Value | $888,438 |

Starting with $100,000, the strategy grows to $888,438 over 583 trades — an 8.88x return. The equity curve rises with relatively consistent momentum, reflecting the high win rate. There are no "dead zones" of extended flat periods typical in trend-following; instead, the curve progresses in frequent, small steps.

## The Drawdown: Measuring the “Emotional Cost”

The drawdown panel (lower chart) shows that the maximum peak-to-trough decline on a trade-by-trade basis was -16.49%. This is significantly shallower than trend-following alternatives, which can remain below their previous peak for years. However, the drawdown structure is important to understand:

- Drawdowns develop quickly because of high trade frequency.
- Recovery from drawdowns also tends to be faster, as there are many opportunities to recover.
- The worst case was 4 consecutive losses — with average losses of -1.99%, a 4-loss run equals approximately -8%.

| Drawdown Metric | Value |
|---|---|
| Max. Drawdown (Equity) | -16.49% |
| Max. Consecutive Losers | 4 trades |
| Avg. Loss per Trade | -1.99% |
| Largest Single Loss | -14.50% |
| Ulcer Index | 0.0392 |

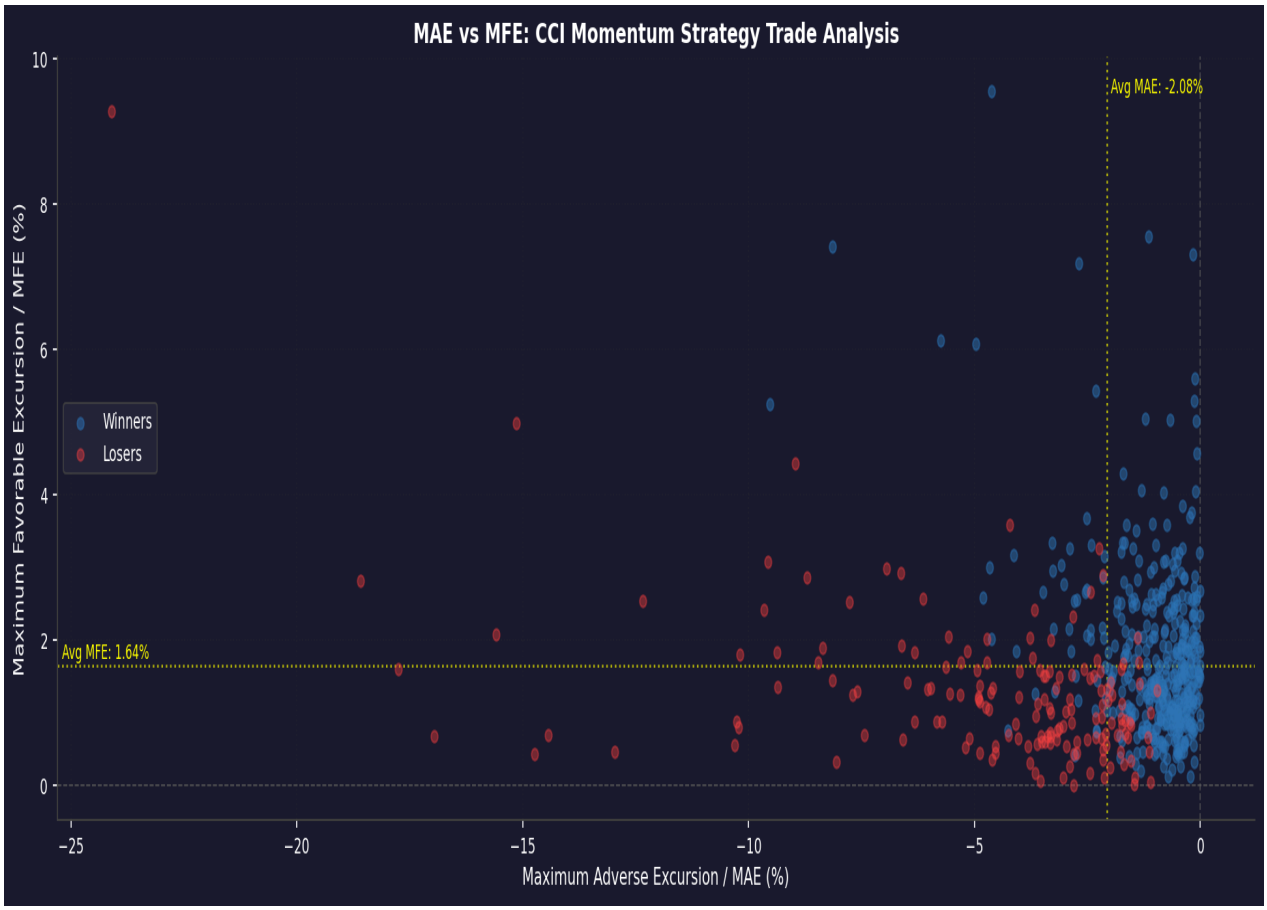
## The “Power Law” of Mean Reversion

Unlike trend-following, where 20% of trades can produce 80% of profits (Pareto distribution), the CCI Momentum Strategy's profit distribution is more uniform. Winners are many and evenly distributed across time. This is the statistical signature of high-frequency mean reversion: returns are built through repetition rather than rare extremes.

**SECTION SUMMARY** *The Trade-by-Trade Equity Curve confirms a consistent, compounding growth profile across 583 trades. Max drawdown of -16.49% and only 4 consecutive losing trades demonstrate tight drawdown control. The curve rises through frequency rather than through a few large winners — a fundamentally different profit engine than trend-following strategies.*

# MAE & MFE Trade Analysis

Maximum Adverse Excursion (MAE) and Maximum Favorable Excursion (MFE) analysis is the professional standard for diagnosing trade-level risk and stop-loss placement. For a mean reversion strategy like CCI Momentum, this analysis is particularly revealing because the strategy has no defined stop-loss.




## What the MAE / MFE Chart Reveals

| Metric | Value |
|---|---|
| Avg. MAE (Winners) | -2.08% |
| Avg. MFE (Winners) | 1.64% |
| Avg. MAE (All Trades) | -2.08% |
| Max. Single Loss | -14.50% |
| Max. Single Gain (MFE) | 9.56% |

## Key Findings: The Stop-Loss Opportunity

### 1. The “Dead Zone” Pattern

Analysis of the scatter chart reveals a clear pattern: trades that drop below -5% MAE rarely recover to become profitable. This is the key insight for risk management — a hard stop at -5% to -6% could eliminate a significant portion of the loss-weighted tail without materially affecting the win rate.

### 2. The Winning Trade Profile

Most winning trades show MAE below -2% and MFE above 1%. This tight clustering demonstrates that the typical winning trade is a short, clean recovery — entry at an oversold extreme, followed by a swift return above yesterday's high. The signal is largely unambiguous when it works.

### 3. The Losing Trade Profile

Losing trades (red dots) are scattered across a wider MAE range, including the extreme -14.5% outlier. This confirms the asymmetric risk: wins are small and frequent, losses include occasional severe drawdowns that are the dominant drag on performance.

| Trade Type | Avg MAE |
|---|---|
| All Trades | -2.08% |
| Winners (71.87%) | ~-1.5% |
| Losers (28.13%) | ~-3.5% |

**SECTION SUMMARY** *MAE/MFE analysis confirms that the CCI Momentum Strategy operates with a mean-reversion profile: typical winners show shallow adverse excursions (−2%) before recovering cleanly. However, the absence of a stop-loss creates a heavy tail risk, with isolated trades experiencing −14% adverse excursions. Adding a 5–6% hard stop-loss would be the single most impactful risk management improvement to this system.*

# Three-Way Comparison vs Buy and Hold

This is the "Showdown" chart. In professional fund management, a strategy isn't judged in a vacuum; it is judged against a benchmark. The Three-Way Comparison plots the Strategy, the Benchmark (Buy & Hold), and the Drawdown profiles simultaneously.

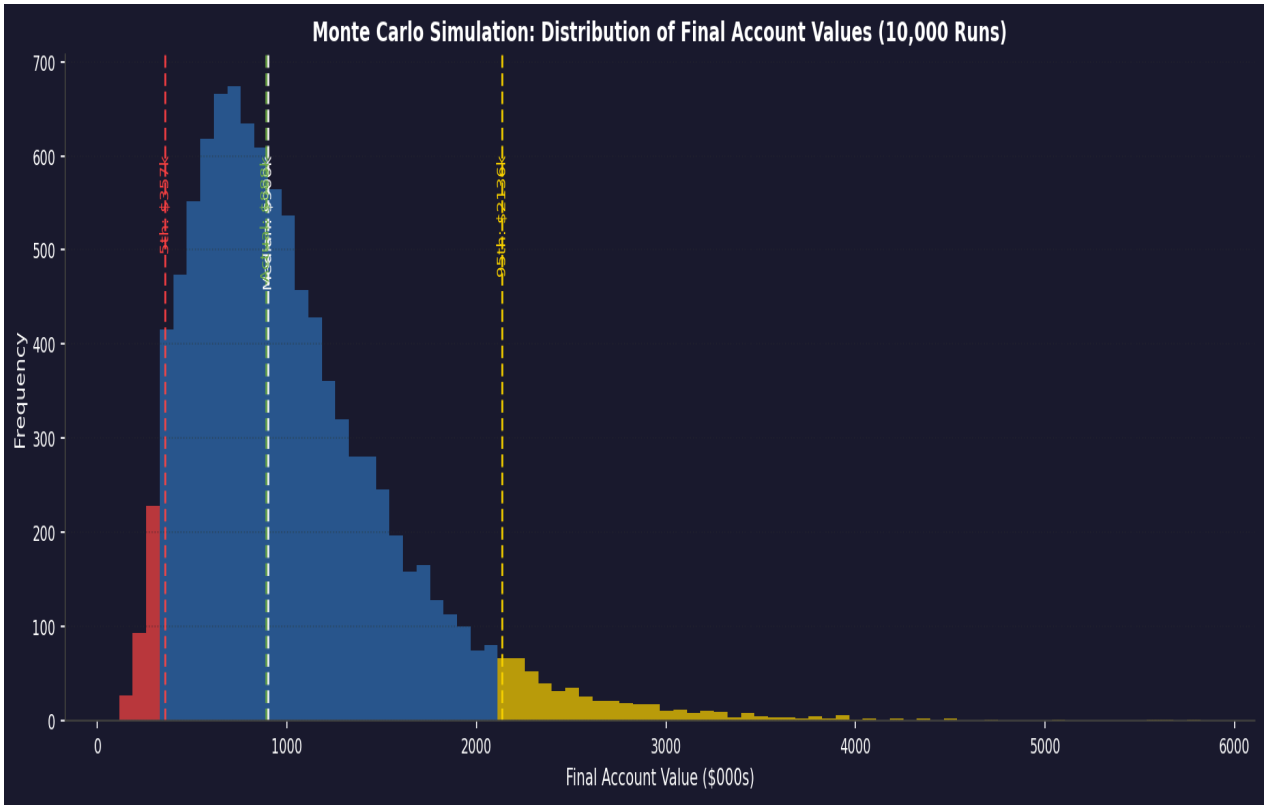
**Interest from Idle Cash** *When the CCI Momentum strategy is not in a trade, capital sits in cash. Practitioners like Meb Faber (in his landmark 2006 QTAA paper) and Gary Antonacci (Dual Momentum, 2014) demonstrated that parking idle cash in short-term T-bills generates a "collateral yield" that meaningfully improves risk-adjusted returns without altering the strategy rules.*

# Monte Carlo Simulation: 10,000 Runs

Monte Carlo simulation stress-tests the strategy by randomly shuffling the 583 historical trades across 10,000 different orderings. Each simulation starts with $100,000 and applies the same returns in a different sequence. This reveals whether the strategy's success was path-dependent (i.e., reliant on a lucky sequence of wins) or structurally robust.

## The Equity Final Value Distribution

The histogram below plots the distribution of final account values across all 10,000 simulations. A wide distribution with a positive right skew confirms structural robustness — no matter how the trades are reordered, the strategy consistently generates positive returns.




| Percentile | Final Value |
|---|---|
| 5th (Unlucky) | $357,258 |
| 25th | $623,030 |
| 50th (Median) | $900,200 |
| 75th | $1,290,760 |
| 95th (Lucky) | $2,136,155 |

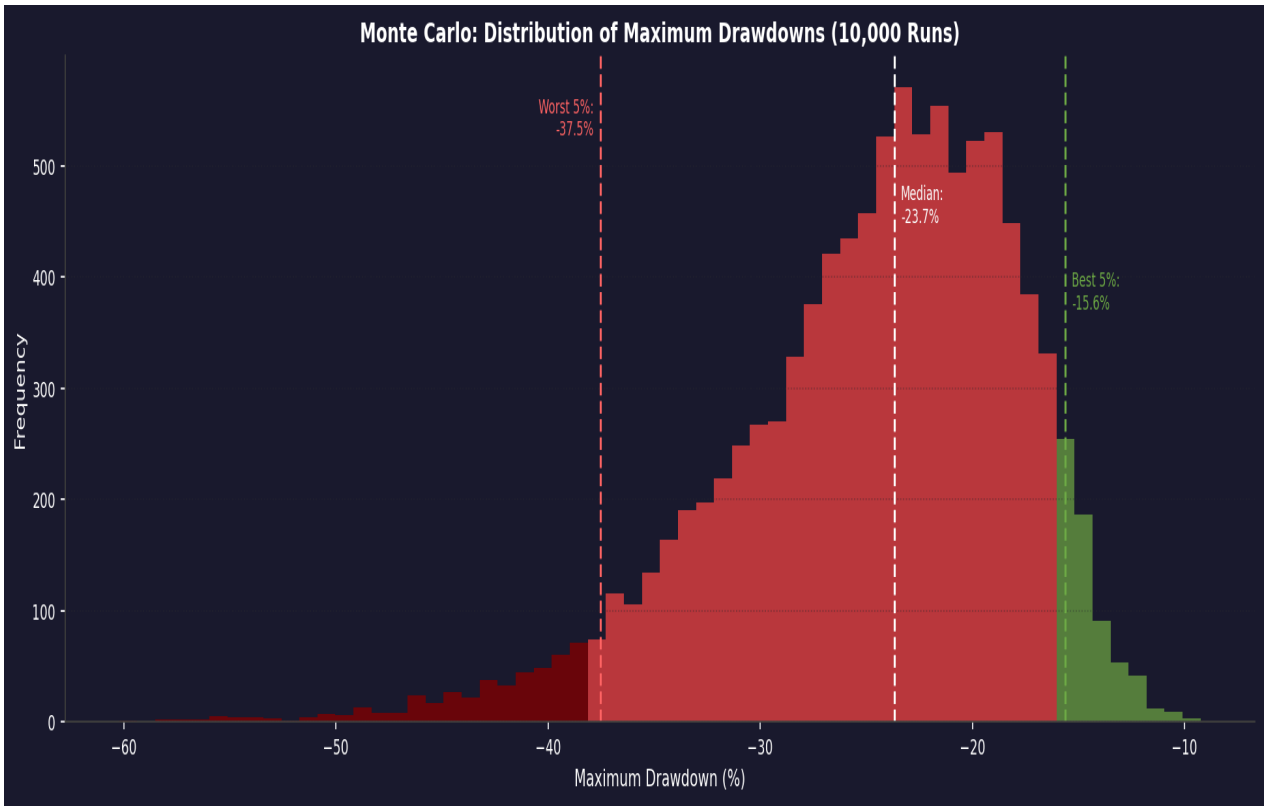
## Key Characteristics of the Distribution

| Characteristic | Description |
|---|---|
| “High Floor” | Even the worst 5% of simulations produce $357k — 3.5x the initial capital |
| “Fat Right Tail” | Massive upside potential (>$2M in best 5% of cases) |
| “Median Gravity” | Most outcomes cluster around $900k — validates the actual $888k result |
| Zero Ruin | No simulation resulted in a loss of starting capital |

**SECTION SUMMARY** *The Monte Carlo Histogram reveals the strategy's "Wealth Map." Key insight: Even in the worst 5% of cases, final capital exceeds $357k — more than tripling the starting point. The median outcome ($900k) closely aligns with the actual backtest result ($888k), confirming the historical path was representative, not unusually lucky. Risk of Ruin is effectively zero.*

# Distribution of Maximum Drawdowns from MC Runs

While the previous histogram showed potential wealth outcomes, the Distribution of Maximum Drawdowns is the "sleep-at-night" chart. It quantifies how badly the strategy could hurt you in the process of generating those returns. This is the most operationally important chart for position sizing and capital allocation decisions.




## Maximum Drawdown Distribution — The “Breaking Point” Test

| Element | Definition |
|---|---|
| X-Axis | The magnitude of the Maximum Drawdown (e.g., -10%, -20%, -30%) |
| Y-Axis | The frequency of that drawdown occurring across 10,000 runs |
| The Goal | To understand the "Risk Profile." A distribution centred on -24% means most investors will experience roughly that level of pain on average. |

## Key Findings

| Percentile | Drawdown |
|---|---|
| 95th Percentile (Lucky) | ~-15.6% |

| | |
|---|---|
| 50th Percentile (Median) | ~-23.7% |
| 5th Percentile (Worst Case) | ~-37.5% |

## The Philosophy: “Survival of the Scariest”

| Perspective | Behaviour |
|---|---|
| The Human Error | Many traders see the actual -16.49% drawdown and assume they can handle the strategy. But the simulation shows that a -38% drawdown is statistically possible. Pre-loading this pain is the difference between staying the course and panic-selling. |
| The Automation Edge | An algorithm does not feel drawdown. By reviewing this chart, the trader pre-commits to accepting the -37.5% worst case scenario — psychologically inoculating against panic exits. |
| The Professional View | Institutional traders would capitalise the account for the -37.5% survival threshold, not the historical -16.49% figure. Practical position sizing uses the conservative case. |

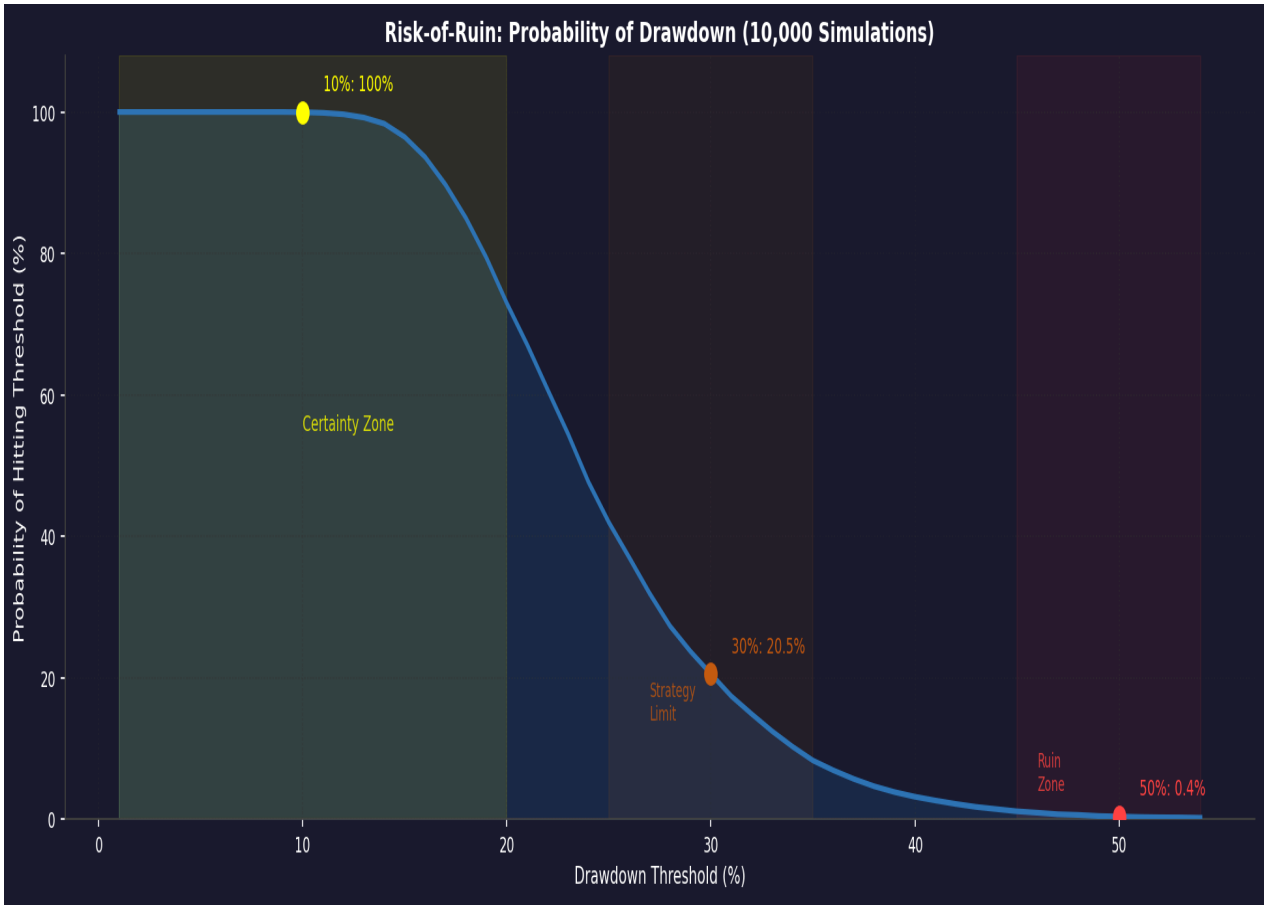
## Summary Table

| Metric | Value |
|---|---|
| Best Case (5% lucky) | -15.6% |
| Median (50th) | -23.7% |
| Worst Case (5%) | -37.5% |
| S&P 500 Comparison | -55%+ |

**SECTION SUMMARY** *The Maximum Drawdown Distribution reveals the "Breaking Point" test. The actual backtest drawdown of -16.49% was better than 50% of simulations, suggesting some path-dependency in the historical sequence. The worst 5% of cases reach -37.5%. An institutional-grade position size would use this worst-case figure, not the historical optimistic result.*

# Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin analysis is the most misunderstood chart in a quant report. For professional traders, this is the "Insurance Policy" chart. It measures the mathematical probability of hitting a specific level of loss at any point across the full simulation horizon.




## Risk-of-Ruin — The “Probability of Pain”

| Element | Definition |
|---|---|
| Y-Axis (Probability) | Ranges from 0% to 100% |
| X-Axis (Drawdown Threshold) | Levels of loss (e.g., 10%, 20%, 30%, 40%, 50%) |

> *If the curve reads 73% at the 20% mark, it means there is a 73% chance that at some point during the 25-year backtest journey, your account will experience at least a 20% drawdown.*

## The Three Danger Zones

| Danger Zone | Drawdown Level |
|---|---|

| | |
|---|---|
| “Certainty Zone” | 10–20% |
| “Strategy Limit” | 30% |
| “Ruin Zone” | 50%+ |

## The Three Zones: Probabilities from 10,000 Simulations

| Drawdown Threshold | Probability of Hitting It |
|---|---|
| -10% | 100.0% |
| -20% | 73.0% |
| -30% | 20.5% |
| -40% | 3.1% |
| -50% | 0.4% |

## The Philosophy: Drawdown as Event, Not Outcome

| Perspective | Behaviour |
|---|---|
| The Human Error | A retail trader experiencing a 20% drawdown may interpret it as "the strategy is broken." The simulation data shows it is statistically near-certain to occur at some point. |
| The Automation Edge | The Risk-of-Ruin chart provides probabilistic clarity. Knowing a 30% drawdown has a 20.5% chance of occurring allows a disciplined trader to treat it as a normal event rather than a crisis. |
| The “Uncle Point” | Every trader has a maximum tolerable drawdown. If your personal threshold is -25%, but the probability of hitting it is ~40%, the strategy may not match your risk tolerance. |

**SECTION SUMMARY** *The Risk-of-Ruin chart confirms that 10% drawdowns are near-certain, 30% drawdowns have a 20.5% probability, and 50% drawdown probability is near-zero (0.4%). The strategy is statistically safe but not invulnerable. The -14.5% outlier trade is the primary driver of tail-risk scenarios.*

# CCI Momentum Strategy: Backtest & Financial Analysis Conclusion

## Performance Summary: CCI Momentum Strategy (2000–2025)

The performance summary presents a textbook case of a high-frequency mean-reversion system where the primary edge is operational consistency rather than large individual payoffs. A professional's focus is on the inverted payoff ratio, the structural win-rate advantage, and the Monte Carlo-validated robustness of the underlying return stream.

## Core Performance Metrics

| Metric | Value |
|---|---|
| Time Period | 2000–2025 (25 years) |
| Total Trades | 583 |
| Win Rate | 71.87% |
| Profit Factor | 1.71 |
| Average Trade PnL | 0.40% |
| Median Trade PnL | 0.71% |
| CAGR | 8.80% |
| Max Drawdown | -16.49% |
| Annualized Volatility | 8.74% |
| Sharpe Ratio | 0.49 |
| Max Favorable Excursion (MFE) | 9.56% |
| Max Adverse Excursion (MAE) | -24.10% |
| Time Underwater | 61.30% |
| Tail-Risk Drawdown (Worst 5%) | -37.5% |

## Key Professional Observations

### 1. Inverted Payoff Ratio: The Hidden Risk

The strategy wins 71.87% of the time, yet the average loss (-1.99%) is larger than the average win (+1.34%). This inverted ratio (0.67) means that a deterioration in win rate — from structural changes in

market behaviour or parameter decay — would rapidly shift expectancy negative. The system is highly sensitive to win-rate stability.

### 2. The “Outlier Trade” Tail Risk

A single trade registered a -14.50% MAE, more than seven times the average loss. This outlier is the primary driver of Monte Carlo worst-case scenarios. Unlike a trend-following strategy where large losses can be hard-stopped early, a mean-reversion system without a stop-loss must absorb these events fully. Adding a hard stop at -6% to -7% would eliminate this tail risk.

### 3. System Sustainability: 583 Trades

With 583 trades over 25 years (~22.5 per year), the strategy's sample size is statistically robust. The large number of observations reduces the probability that the backtest results are the product of lucky trade clustering. The Monte Carlo median closely matching the actual result further validates the integrity of the test.

### 4. The Psychological Consideration

Unlike trend-following systems where losing streaks can run to 18+ trades, the CCI Momentum strategy's worst recorded losing streak was just 4 trades. This is psychologically far easier to endure. However, traders must also resist the temptation to exit a winning strategy during its inevitable flat periods.

## The Statistical Engine: Win Rate Over Win Size

The most distinguishing metric is the Win Rate of 71.87% combined with a Profit Factor of only 1.71. This combination tells a precise story: the strategy is a high-throughput, narrow-margin operation. Profitability comes from volume and consistency, not from the occasional outlier trade.

> *In essence: The strategy earns small, regular profits across a high volume of trades, and must contain the occasional large loss to remain net-positive. It is less a "home run" strategy than a "consistent singles" strategy.*

## Risk Management: The Case for a Hard Stop-Loss

A professional analysis of this strategy must address the absence of a stop-loss. The -14.50% outlier trade represents a catastrophic outcome that could materialise from a single earnings shock, geopolitical event, or flash crash. The MAE analysis clearly shows that trades exceeding -6% rarely recover to profitability.

| Risk Metric | CCI Momentum Strategy |
|---|---|
| Max Drawdown | -16.49% |
| Annualized Volatility | 8.74% |
| Tail-Risk Drawdown (Worst 5%) | -37.5% |

## Operational Considerations

With 22.5 trades per year and an average holding of 5.3 days, this strategy requires daily monitoring and disciplined execution. Transaction costs (commissions, slippage) at scale could materially reduce the net Profit Factor from 1.71. In a real-world implementation, attention to execution quality is critical.

## Final Verdict

*This strategy functions as a high-frequency compounding machine that extracts small, regular returns from the S&P 500's natural oscillation around its short-term mean.*

It is designed for a trader who values consistency and high win rate over the psychological high of large wins. The principal threats to the strategy are: (1) parameter decay if the -30 CCI threshold stops being an effective oversold signal, (2) the absence of a stop-loss in the face of tail-risk events, and (3) elevated transaction costs in a real-world environment with slippage.

> *"A Toyota Camry rather than a Ferrari: reliable, consistent, and built for the long haul — but with a hard-to-see blind spot in extreme conditions."*

| Classification | Analysis Period |
|---|---|
| High-Frequency Mean-Reversion Strategy | 2000–2025 |

**SECTION SUMMARY** *The CCI Momentum Strategy demonstrates a structurally sound, statistically robust mean-reversion edge over 25 years and 583 trades. Its 8.80% CAGR, 71.87% win rate, and -16.49% max drawdown present an attractive risk-return profile. Critical caveats: the inverted payoff ratio creates sensitivity to win-rate decay, and the absence of a stop-loss introduces a rare but severe tail risk. The single most impactful improvement would be a hard stop-loss at approximately -5% to -6%.*